

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^∞ norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Thought for 230 seconds >

$$x_1 = 0.00, x_2 = 1.00$$

Explanation:

The computable number a_2 is defined as the limit of the given algorithm as $n \rightarrow \infty$.

The function $f(n)$ sums $(1/2)^k$ for $k = 3, \dots, n+2$ and adds/subtracts terms of order $72^{\{-10^{24}\}}$. As $n \rightarrow \infty$, the sum of $(1/2)^k$ tends to $1/4$, and the $72^{\{-\dots\}}$ terms are astronomically

tiny and positive. Thus $a_2 = 2 \cdot (1/4 + \varepsilon) + 1/2 = 1 + 2\varepsilon$ with $\varepsilon > 0$ but unimaginably small ($\varepsilon \approx 72^{-10^{24}}$).

We want to minimise total time $T = x_1 + x_2$ subject to $a_1x_1 + a_2x_2 = 1$, $x_1, x_2 \geq 0$, with $a_1 = 1$ and $a_2 > 1$.

Since $a_2 > 1$, the optimal is to use only the faster therapy: set $x_1 = 0$, then $x_2 = 1/a_2 \approx 1$.

To any practical accuracy, $a_2 = 1$ and the unique minimiser is $(0, 1)$. Rounded to two decimal places, this gives $x_1 = 0.00$ and $x_2 = 1.00$.